

The Distance Formula and the Midpoint in Two Dimensions (Homework).

Find the distance between the following coordinates.

Answers

① $(3, -2)$ & $(-4, 1)$

$\sqrt{58}$

② $(1, 1)$ & $(-2, -5)$

$\sqrt{45} = 3\sqrt{5}$

③ $(-6, -1)$ & $(-2, 2)$

5

④ $(0, 3)$ & $(2, 1)$

$\sqrt{8} = 2\sqrt{2}$

⑤ $(4, 0)$ & $(0, 0)$

4

⑥ $(-8, -1)$ & $(-2, -3)$

$\sqrt{40} = 2\sqrt{10}$

⑦ $(3, -1)$ & $(0, 7)$

$\sqrt{73}$

Find the midpoint between the following coordinates.

⑧ $(10, -3)$ & $(2, -1)$

$(6, -2)$

⑨ $(-2, 0)$ & $(5, 2)$

$(\frac{3}{2}, 1)$

⑩ $(7, 4)$ & $(11, 6)$

$(9, 5)$

⑪ $(-4, -2)$ & $(-1, -3)$

$(-\frac{5}{2}, -\frac{5}{2})$

⑫ $(3, 4)$ & $(0, 0)$

$(\frac{3}{2}, 2)$

⑬ $(1, -4)$ & $(-3, 3)$

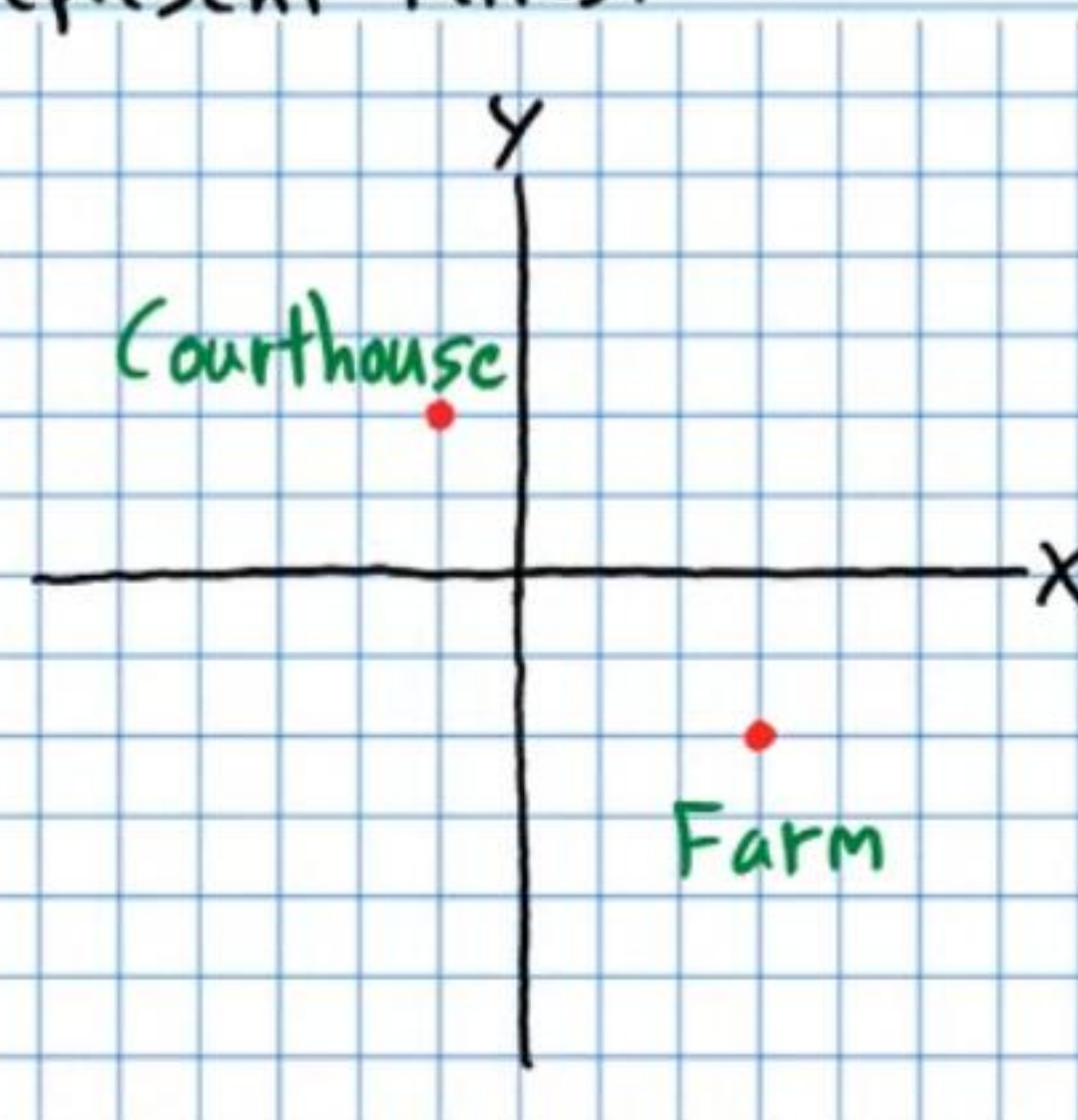
$(-1, -\frac{1}{2})$

⑭ $(-8, 6)$ & $(8, -6)$

$(0, 0)$

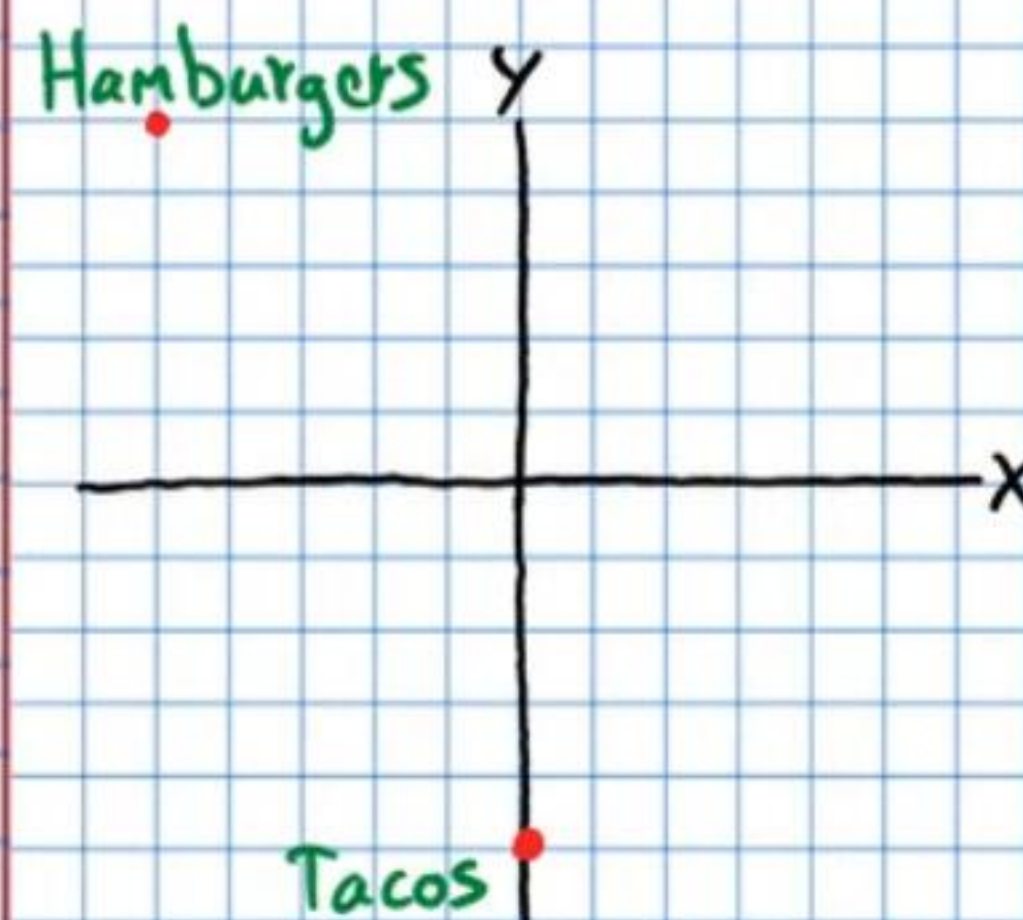
⑮ Find the distance between the farm and the courthouse. The units on the grid below represent miles.

$\sqrt{32} = 4\sqrt{2} \text{ mi}$

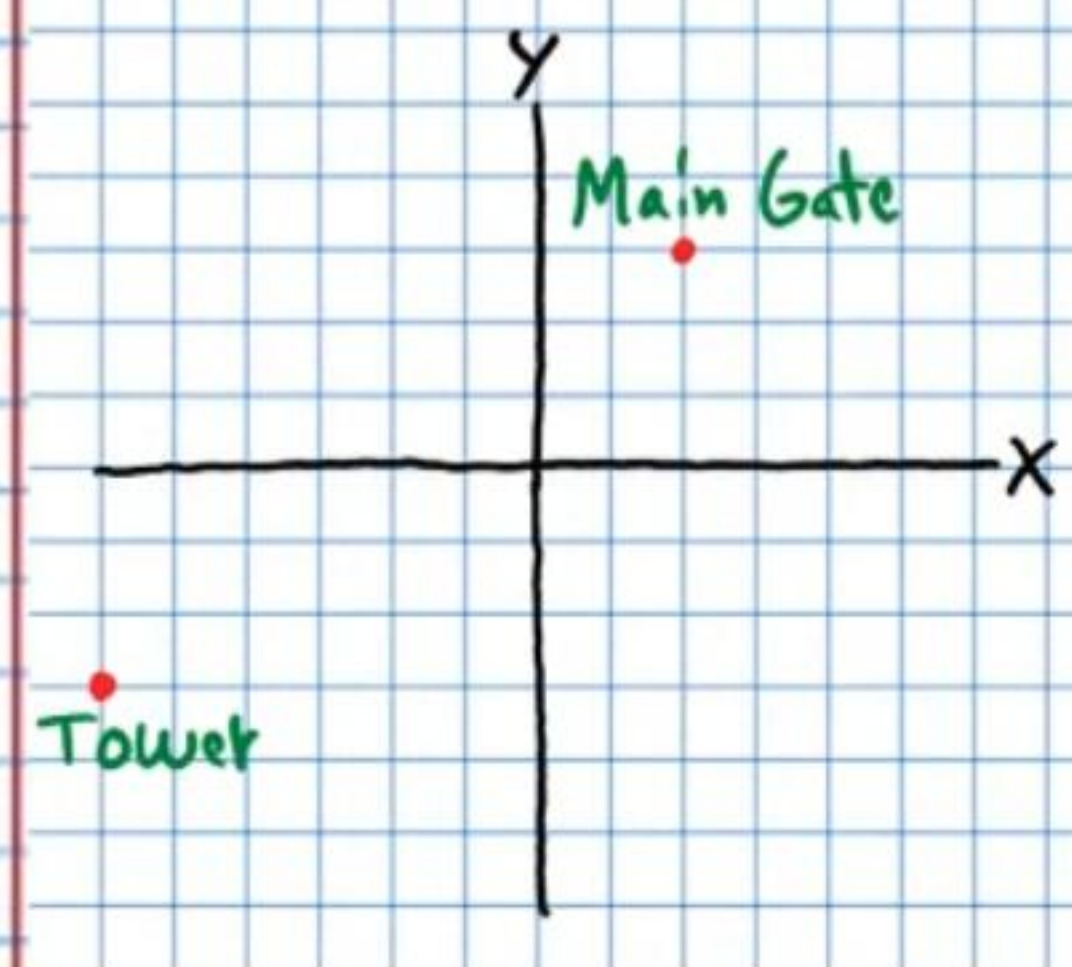


⑯ Find the distance between the hamburger restaurant and the taco restaurant. The units on the grid represent kilometers.

$\sqrt{125} = 5\sqrt{5} \text{ km}$

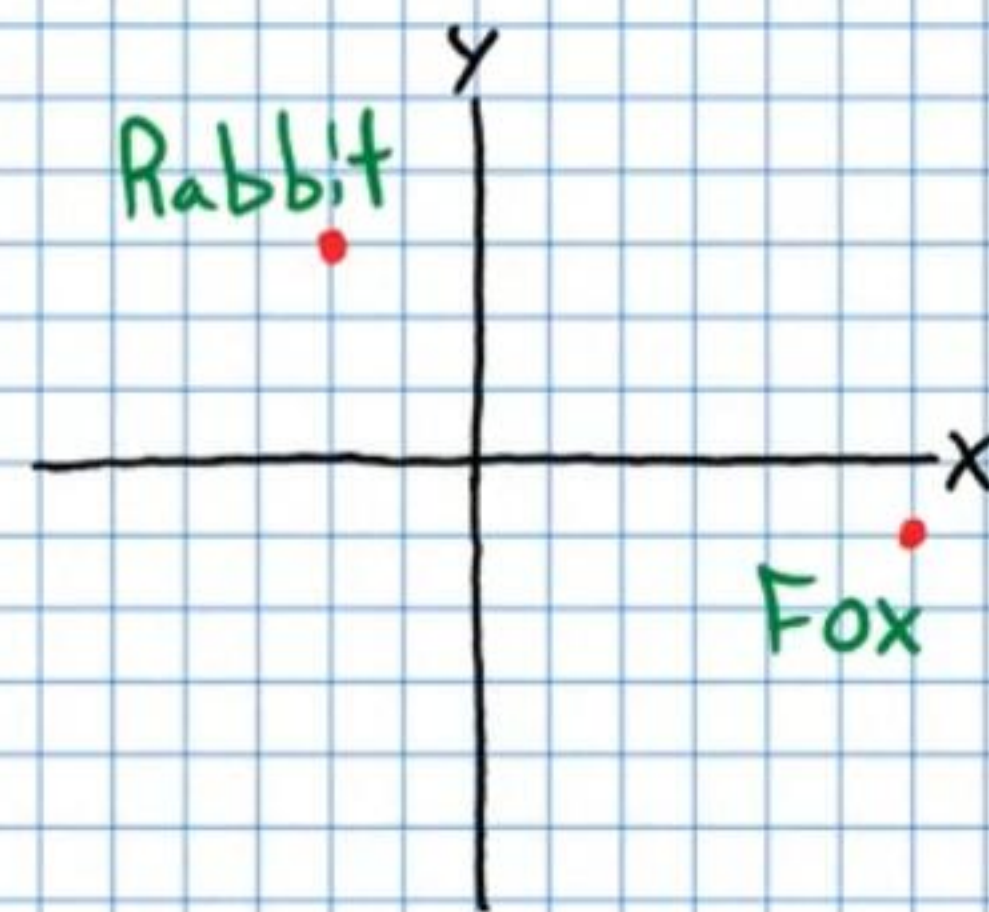


⑪ Find the distance between the tower and the main gate. Each unit on the grid represents six kilometers.



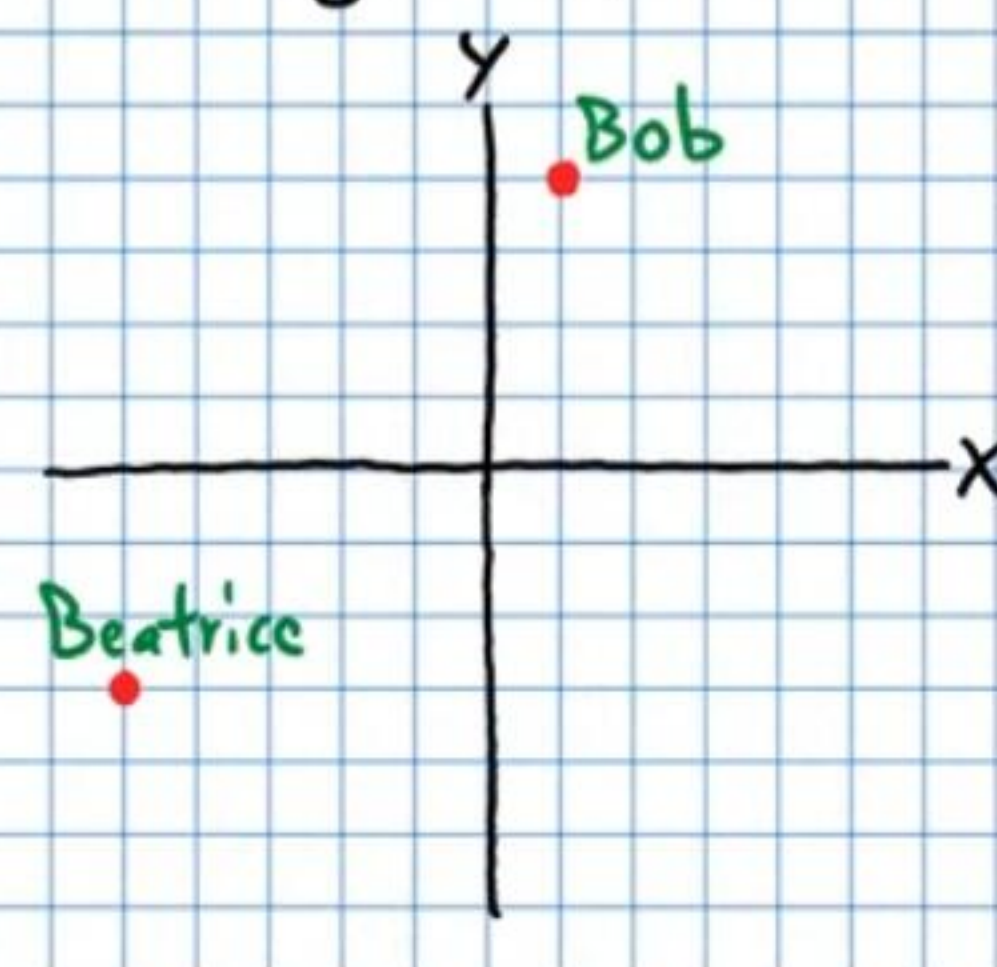
$$60 \text{ km}$$

⑪ Find the distance between the rabbit and the fox. Each unit on the grid represents two meters.



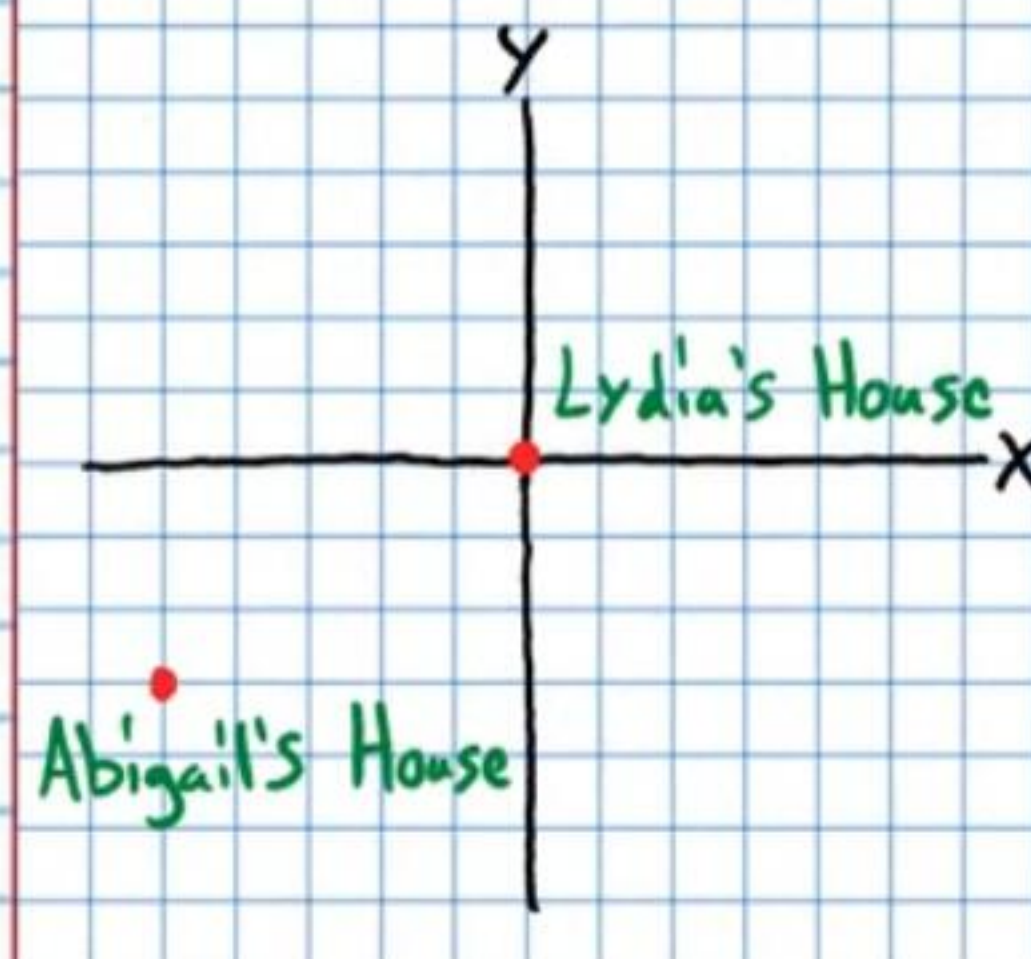
$$8\sqrt{5} \text{ m}$$

⑱ Bob and Beatrice want to meet halfway for lunch. How far will each of them have to travel? Each unit on the grid represents 200 yards.



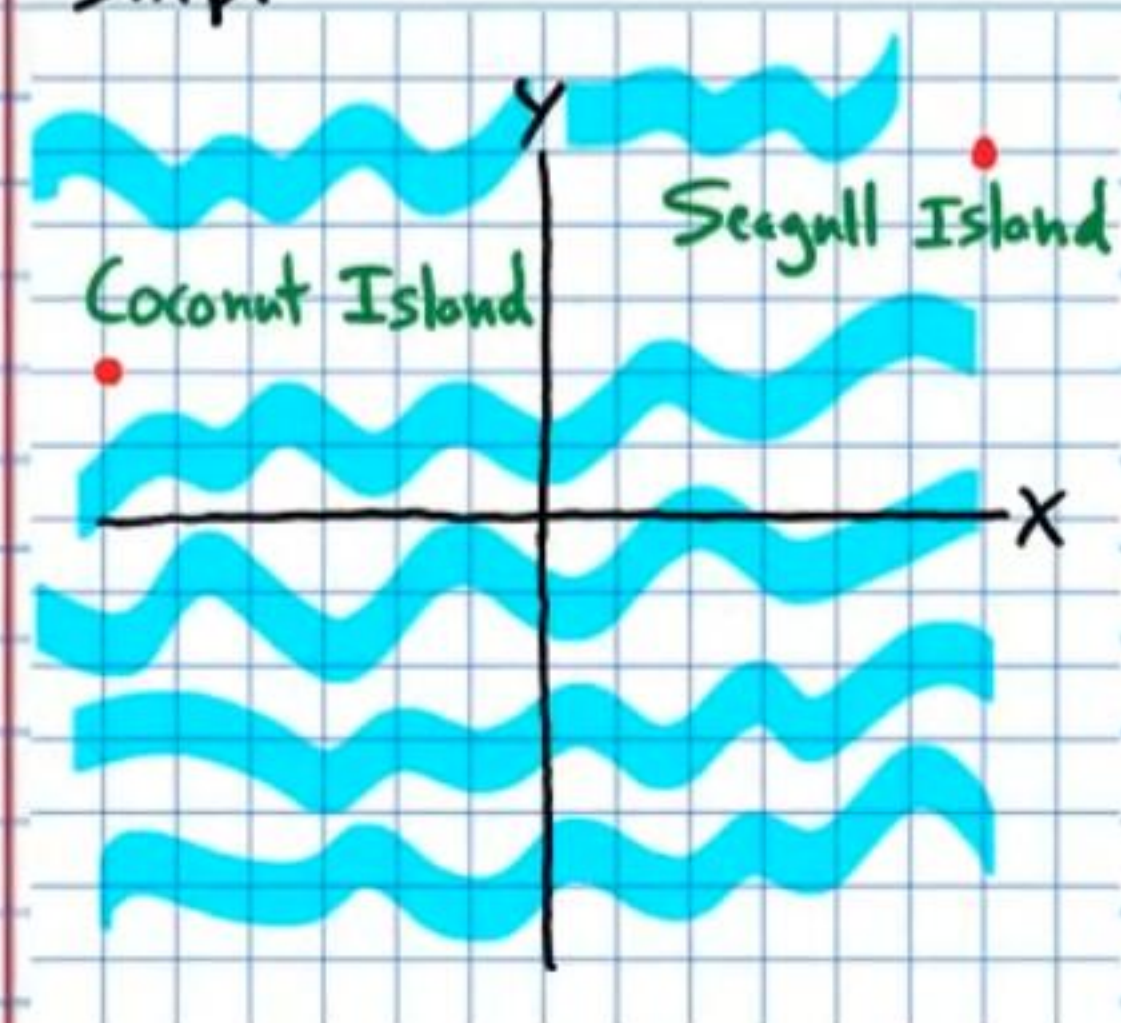
$$100\sqrt{85} \text{ yd}$$

⑳ The mall is halfway between Lydia's house and Abigail's house. What are the coordinates of the mall?



$$\left(-\frac{5}{2}, -\frac{3}{2}\right)$$

- ② A ship is located halfway between Coconut Island and Seagull Island. Find the coordinates of the ship.



$$\left(0, \frac{7}{2}\right)$$

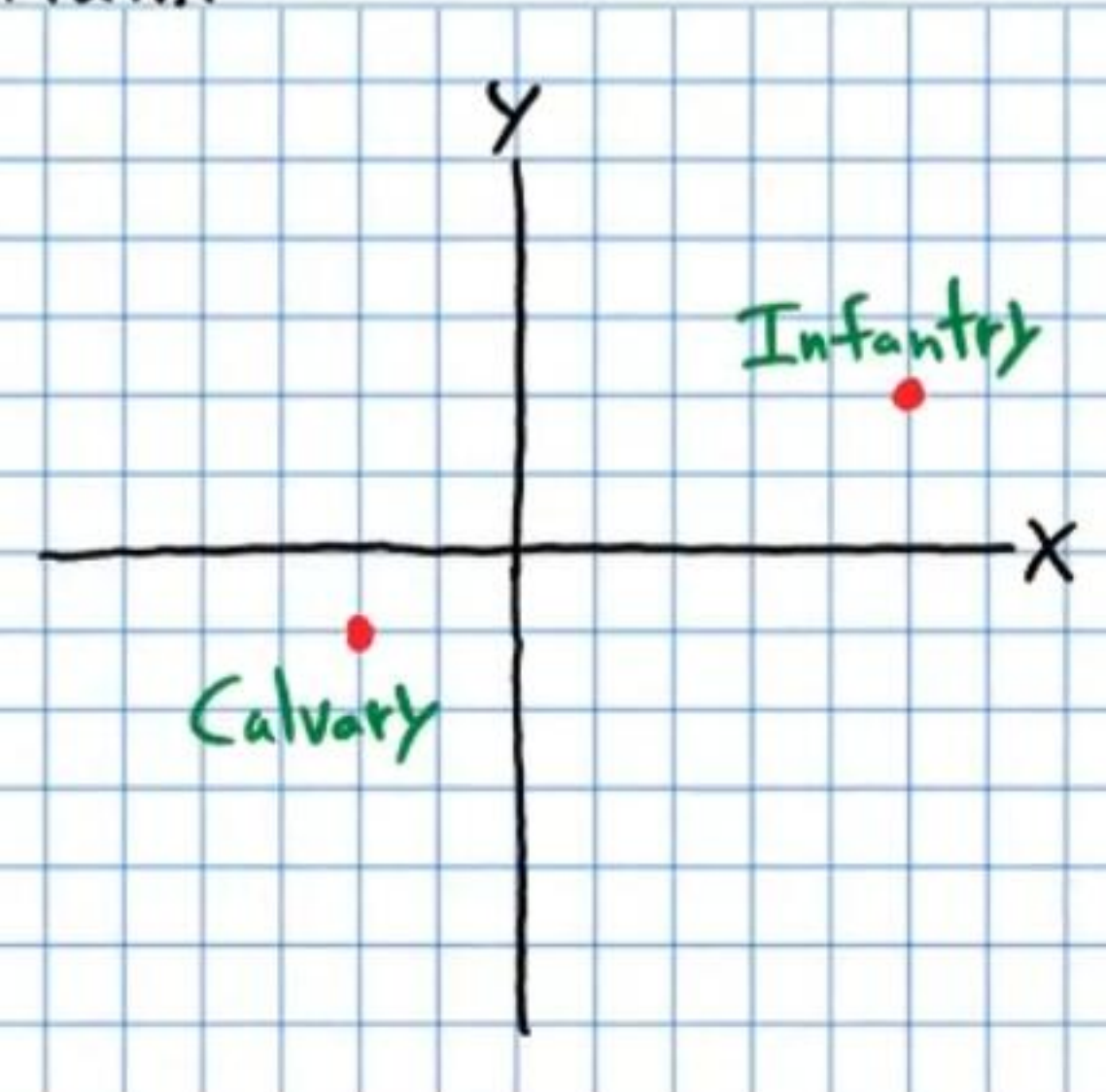
- ② If $(5, 1)$ is the midpoint between points E and F, and the coordinates of E are $(-6, -3)$, find the coordinates of F.

$$F(16, 5)$$

- ③ If $(-8, 9)$ is the midpoint between points Q and R, and the coordinates of R are $(3, 0)$, find the coordinates of Q.

$$Q(-19, 18)$$

- ④ The cavalry is halfway between the infantry and the supply train. Find the coordinates of the supply train.



$$(-9, -4)$$

- ⑤ The distance between $(3, -4)$ and $(X, 1)$ is $\sqrt{89}$. Find the possible values of X.

$$11 \text{ or } -5$$

- ⑥ The distance between $(-2, y)$ and $(2, 4)$ is $\sqrt{17}$. Find the possible values of y.

$$3 \text{ or } 5$$